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## Operating instructions

**CE** The unit complies with the corresponding EC guidelines.

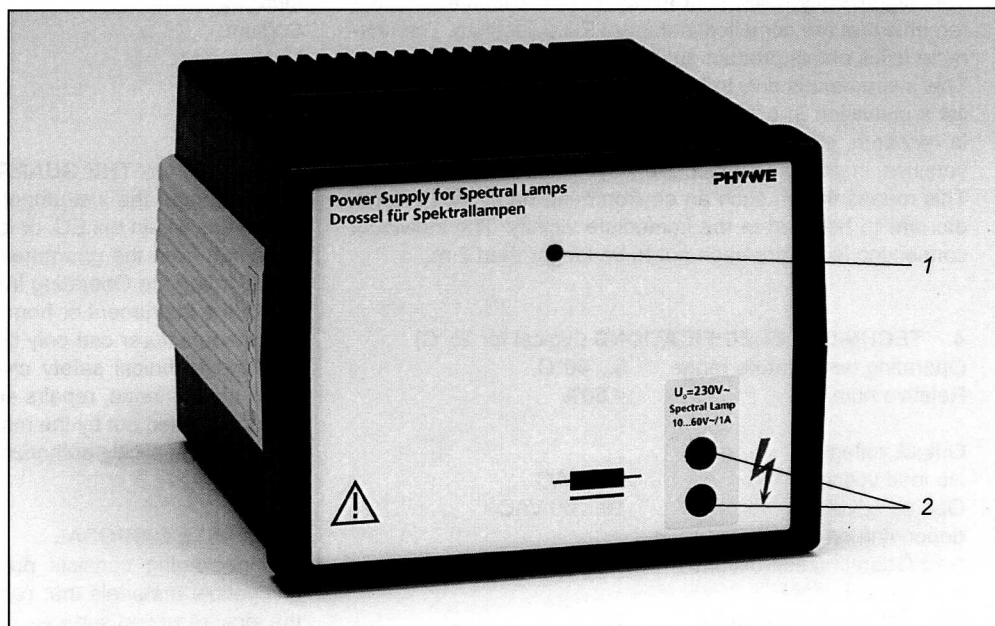


Fig. 1: Front view of the Power supply for spectral lamps 13662.97.

## 1 SAFETY PRECAUTIONS



- Carefully read these operating instructions before operating this instrument. This is necessary to avoid damage to it, as well as for user-safety.
- Check that your mains supply and mains frequency voltage corresponds to that given on the type plate fixed to the instrument.
- Install the instrument so that the on/off switch and the mains connecting plug are easily accessible.
- Do not cover the ventilation slots.
- Take care that no liquids or objects enter in through the ventilation slots.
- Only use the instrument in dry rooms in which there is no risk of explosion.
- Do not start up this instrument should there be visible signs of damage to it or to the line cord.
- Only use the instrument for the purpose for which it was designed.

## 2 PURPOSE AND DESCRIPTION

The power supply is used for spectral lamps of the 08120.01... 08120.14 series (with Pico 9 socket).

A pair of 4 mm sockets on the front side is used to connect a socket suited for spectral lamps. To ignite the spectral lamp, the power supply provides a no load voltage  $U_0 = 230$  VAC,

which decreases to the corresponding operating voltage  $U_B = 10... 60$  VAC once the lamp is ignited.

For safety reasons, both output sockets are designed as switching sockets, so that the voltage only is available at the output once both plugs have been pushed into the sockets. This excludes all risks for the persons carrying out experiments.

The unit is accommodated in an impact resistant plastic housing. A retractable carrying handle is recessed into the top cover of the unit. The same component is recessed into the bottom panel and can be folded out to enable the unit to be used in a sloped position. Four rubber feet ensure that the unit stands firm and they also enable the it to be stacked with units of a similar design, the rubber feet being located in the cup-shaped depressions on the lower unit to stop the upper unit slipping. When units are stacked, the sloped position is only permissible for the topmost unit.

The unit is connected to the AC mains using the mains lead that is supplied with it. The lead is inserted into the equipment connection plug at the back of the unit. Please ensure that the mains voltage matches the operating voltage details on the nameplate on the unit.

The mains switch to start operating the unit is situated in the immediate vicinity of the equipment connection plug at the back of the unit.

The following functional and operating elements (see Fig. 1) are to be found on the front plate of the instrument:

- 1 On/off indicating lamp
- 2 Output  
Pair of 4 mm sockets for lamp voltage supply

**Changing the fuse:** to change a blown fuse, the fuse holder underneath the main plug must be opened using a screw-driver.

**To change the fuse, the main plug must be disconnected from the grid.**

### 3 NOTES ON OPERATION

This high-quality instrument fulfills all of the technical requirements that are compiled in current EC guidelines. The characteristics of this product qualify it for the CE mark.

This instrument is only to be put into operation under specialist supervision in a controlled electromagnetic environment in research, educational and training facilities (schools, universities, institutes and laboratories).

This means that in such an environment, no mobile phones etc. are to be used in the immediate vicinity. The individual connecting leads are each not to be longer than 2 m.

### 4 TECHNICAL SPECIFICATIONS (typical for 25°C)

Operating temperature range 5... 40°C  
Relative humidity < 80%

Output voltage  
No load voltage 230 VAC  
Operating voltage 10... 60 VAC  
depending on model  
(see Operating instructions  
of Spektral lamps)  
Nominal current limited to 1 A

#### Mains supply

The instrument corresponds to protection class I. It is only to be connected to a socket with an earth lead connection.

Connecting voltage see type plate\*  
(+6%/-10%)  
Mains frequency see type plate\*  
Mains fuse see type plate  
(5 mm x 20 mm)  
Housing dimensions (mm) 230 x 236 x 168 (W, D, H)  
Weight approx. 4.4 kg

\* Voltage and frequency (see type plate) depending on local power grid

xxxxx.91 = 115 V/60 Hz

xxxxx.92 = 115 V/50 Hz

xxxxx.94 = 230 V/60 Hz

xxxxx.97 = 230 V/50 Hz

Special voltages and fixed frequencies on request.

### 5 ACCESSORIES

|                                         |          |
|-----------------------------------------|----------|
| Lamp holder, pico 9, for spectral lamps | 08119.00 |
| Cadmium lamp for Zeeman effect          | 09050.20 |
| Experiment lamp 6                       | 11615.05 |

#### Spectral lamps

|         |          |
|---------|----------|
| Cadmium | 08120.01 |
| Helium  | 08120.03 |
| Mercury | 08120.14 |
| Sodium  | 08120.07 |
| Neon    | 08120.08 |
| Zinc    | 08120.11 |

### 6 NOTES ON THE GUARANTEE

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The manufacturer can only be held responsible for the function and technical safety characteristics of the instrument, when maintenance, repairs and alterations to the instrument are only carried out by the manufacturer or by personnel who have been explicitly authorized by him to do so.

### 7 WASTE DISPOSAL

The packaging consists predominately of environmentally compatible materials that can be passed on for disposal by the local recycling service.



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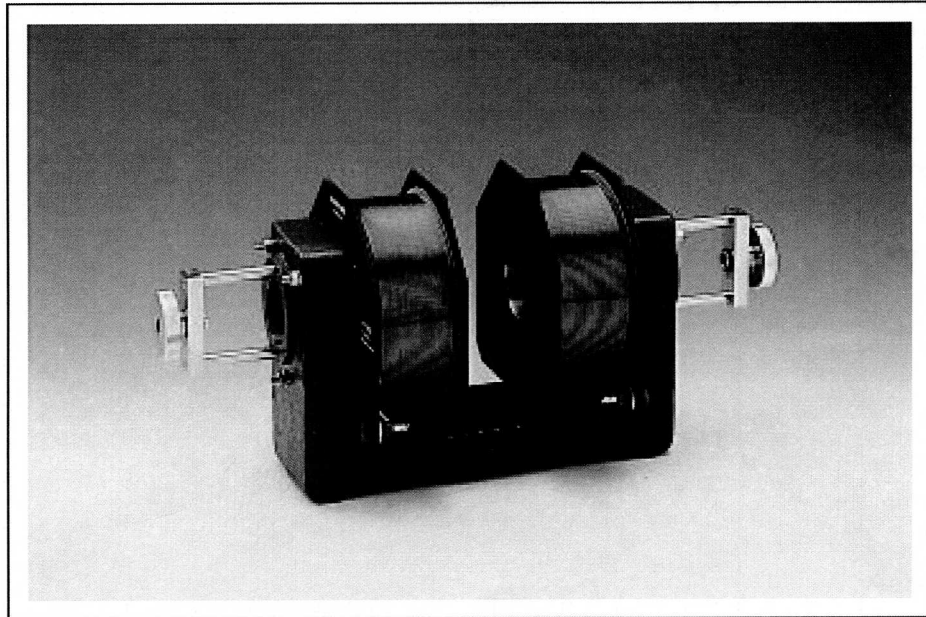
Fax +49 (0) 551 604-246



## Electromagnet with work-locating fixture

06480.01

### Operating Instructions



#### 1. FUNCTION AND DESCRIPTION

The electromagnet 06480.01 is used in conjunction with the pole shoes 06482.02, 06480.03 or 06497.00 to produce strong magnetic fields. The air gap can be varied sensitively with the aid of the worm drive of the work-locating fixture. The electromagnet is particularly suitable e.g. for the Stern-Gerlach experiment, for experiments on diamagnetism and paramagnetism, the Zeeman effect, the Hall effect, and many others.

The magnet consists of a U-shaped iron core with coils which can optionally be connected in parallel or in series. When the coil current is switched off, the field energy in the coil flanges is nullified so that only a slight excess voltage occurs on the coil terminals.

A smoothed direct voltage is not required to operate the electromagnet. As the source of energy we recommend the direct voltage output of the variable-ratio transformer with rectifier 25 V AC/20 V DC, 13531.93.

#### 2. MOUNTING THE POLE SHOES

To mount the pole shoes 06480.02 and 06497.00, loosen both holding screws on the spindle supports of the magnet and remove the support; the locking bolts remain on the magnet.

Each pole shoe is screwed onto a spindle support. The pole shoe is then inserted through the iron core of the magnet coil so that one of the two locking bolts lies between the guide pins of the pole shoe and thus prevents the latter from rotating. The spindle support is finally clamped with the two screws on the locking bolts.

The position of the shoes, and hence the size of the air gap, can be optimized according to the experimental conditions with the aid of the knurled-head screws.

In order to mount the conical, pierced pole shoes 06480.03, the retaining bolts must be removed from the magnet. To loosen the bolts, it is advisable to use universal pliers. The pole shoes are passed through the holes in the iron core of the magnet, until the fixing flanges come against the core. It must then be checked, whether the air gap between the pole shoes is wide enough. To correct the air gap, the clamping screws of the fixing flanges are loosened and the flanges are shifted as required. The clamping screws must then be well tightened, as otherwise the cores might be pulled together due to magnetic forces, which may cause the destruction of parts placed in the air gap.

**Warning!** Owing to the strong magnetic forces, the size of the air gap is reduced slightly when the magnetic field is switched on. Glass apparatus arranged in the air gap can thus be damaged if placed too close to the pole shoes.

Damage such as this can be avoided by switching on the magnetic field before the glass apparatus has been inserted, setting the maximum coil current required for the experiment, and introducing the glass apparatus between the pole shoes only after the current has been switched on.

### 3. TECHNICAL DATA

|                           |                               |
|---------------------------|-------------------------------|
| Coil terminals            | 4 mm sockets                  |
| Number of turns per coil  | 842                           |
| Resistance of each coil   | 2.66 $\Omega$                 |
| Maximum current intensity | 5 A (short-time duty 20 min.) |

Permissible continuous current 4 A

Flux density ( $I = 5$  A) with pole shoes 06480.02:

|              |                    |
|--------------|--------------------|
| Width of gap | 2.5 mm ca. 1.3 T   |
|              | 5.0 mm ca. 1.1 T   |
|              | 10.0 mm ca. 0.7 T  |
|              | 20.0 mm ca. 0.45 T |

|            |                                                |
|------------|------------------------------------------------|
| Dimensions | with 350 mm,<br>depth 140 mm,<br>height 180 mm |
|------------|------------------------------------------------|

|        |           |
|--------|-----------|
| Weight | ca. 17 kg |
|--------|-----------|

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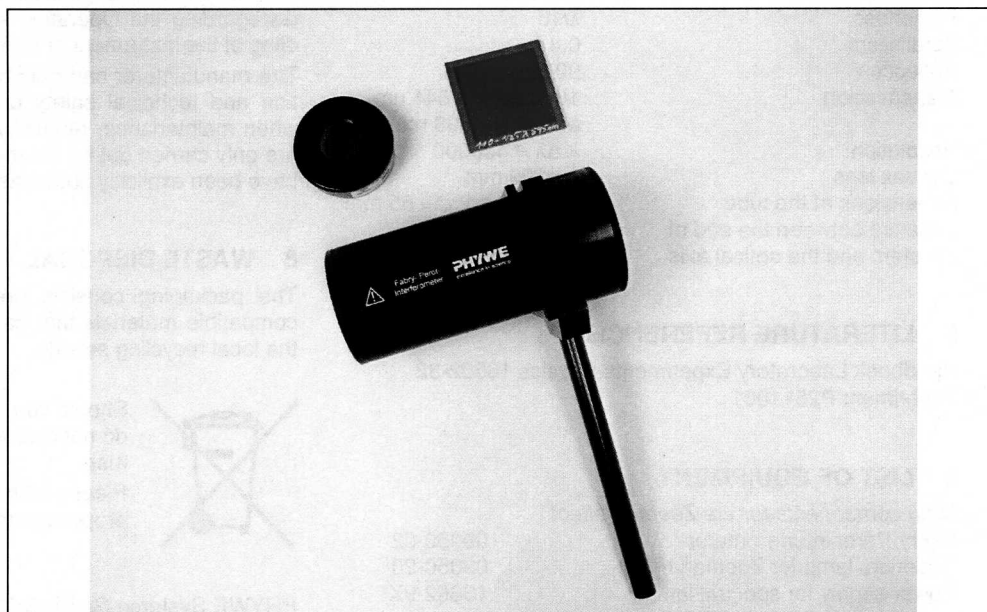


Fig. 1: 09050-03 Fabry-Perot interferometer

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### 1 SAEFTY PRECAUTIONS



#### Caution!

- Carefully read these operating instructions before operating this instrument. This is necessary to avoid damage to it, as well as for user-safety.
- Do not start up this instrument in case of visible signs of damage to it.
- Only use the instrument for the purpose for which it was designed.

### 2 PURPOSE AND CHARACTERISTICS

This high resolution interferometer serves, in combination with the PHYWE cadmium lamp (09050-20), for the observation and measurement of the normal and the anomalous Zeeman effect. The Fabry-Perot etalon is adjusted to two different Cd-lines and provides very intense interference rings that can be observed and measured using optical devices. When magnetic systems with axially bored pole pieces are used, the longitudinal and the transverse Zeeman effects can both be observed.

### 3 FUNCTIONAL AND OPERATING ELEMENTS

The Fabry-Perot etalon consists of a completely flat quartz glass plate that is partially transparent mirrored on both sides. Multiple reflections within the plate lead to interferences. The distance between the mirrored surfaces is exactly 3 mm. A converging lens is firmly mounted in the tube for intense illumination of the etalon. The tube features a slot where a red filter can be inserted between the etalon and the converging lens. The 508 nm interference filter that is supplied can be mounted by a lens holder of type 08012-00. The interferometer is held by the metal rod that can be screwed in the tube from the side.



#### 4 TECHNICAL SPECIFICATIONS

|                                                              |                                                           |
|--------------------------------------------------------------|-----------------------------------------------------------|
| Plate diameter                                               | 35 mm                                                     |
| Plate thickness                                              | 3 mm                                                      |
| Refraction index                                             | $n = 1.45$                                                |
| Planeness                                                    | $\lambda/10$                                              |
| Parallelism                                                  | 0.4 "                                                     |
| Reflection                                                   | 90 %                                                      |
| Transmission                                                 | 10 % (at $\lambda = 644$ nm<br>and at $\lambda = 508$ nm) |
| Resolution                                                   | $\lambda/\Delta\lambda = 400000$                          |
| Convex lens                                                  | $f = 100$ mm                                              |
| Dimensions of the tube                                       | $l = 115$ mm; $d = 55$ mm                                 |
| Distance between the end of<br>the stem and the optical axis | 150 mm                                                    |

#### 5 LITERATURE REFERENCE

Handbook Laboratory Experiments Physics 16502-32  
Experiment P2511001

#### 6 LIST OF EQUIPMENT

##### *Main components for the Zeeman effect*

|                                 |           |
|---------------------------------|-----------|
| Fabry-Perot interferometer      | 09050-02  |
| Cadmium lamp for Zeeman effect  | 09050-20  |
| Power supply for spectral lamps | 13662-9X* |

##### *Magnetic systems*

|                                   |          |
|-----------------------------------|----------|
| Electromagnet without pole pieces | 06480-01 |
| Pole pieces, drilled, conical     | 06480-03 |
| Rotating table for heavy loads    | 02077-00 |

or

|                             |          |
|-----------------------------|----------|
| Magnetic system, adjustable | 06327-00 |
|-----------------------------|----------|

\* Voltage and frequency (see type plate) depending on local power grid.

xxxxx.91 = 115 V/60 Hz

xxxxx.97 = 230 V/50 Hz

#### 7 NOTES ON THE GUARANTEE

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#### 8 WASTE DISPOSAL

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## Cadmium lamp for Zeeman effect

09050.20

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Internet [www.phywe.de](http://www.phywe.de)

### Operating Instructions



The unit complies  
with the corresponding  
EC guidelines.

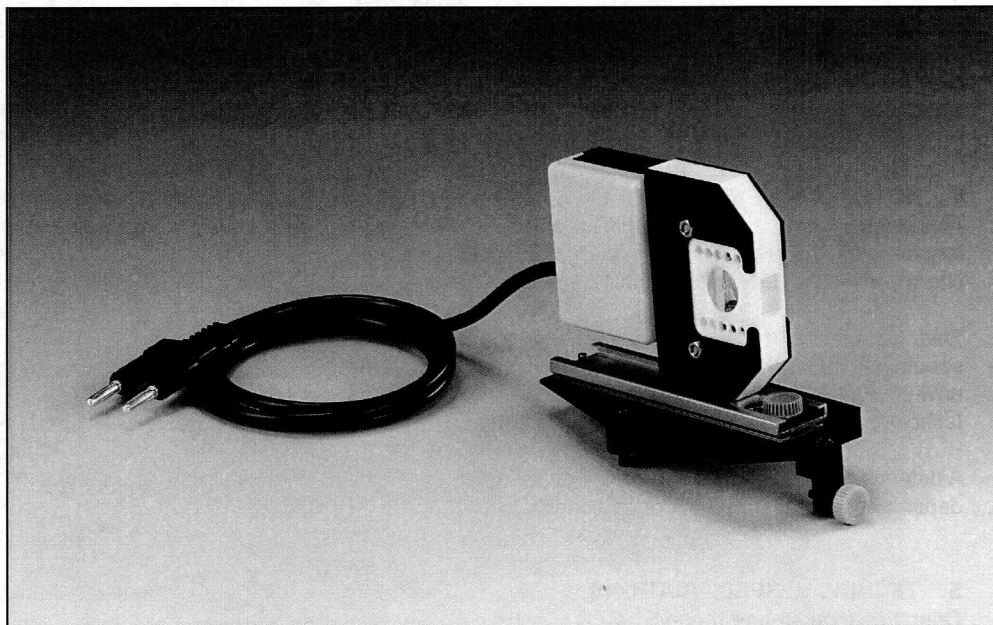


Fig. 1: Cadmium lamp for Zeeman effect 09050.20.

### 1 SAFETY PRECAUTIONS



- Carefully read these operating instructions completely before operating this instrument. This is necessary to avoid damage to it, as well as for user-safety.
- Only use the instrument in dry rooms in which there is no risk of explosion.
- Do not start up this instrument in case of visible signs of damage to it.
- Only use the instrument for the purpose for which it was designed.
- The Cadmium lamp for Zeeman effect is only to be used with the operating unit intended for this purpose (Power supply for spectral lamps, article no. 13662.9X\*).
- **Never connect the two 4 mm plugs of the connecting lead directly to the electric mains (230 V)!**
- Only operate the lamp in the vertical position.
- Only look directly at the lamp through the optical set-up intended for this.

### 2 PURPOSE AND CHARACTERISTICS

The Cd-lamp serves, together with the PHYWE Fabry-Perot interferometer, for the observation and measurement of normal and the anomalous Zeeman effects.

The lamp holder is so designed that it can be mounted either directly between the axially drilled pole pieces of the PHYWE electromagnet (article no. 06480.01) or to the adjustable magnetic system (article no. 06327.00). Adjustment to the different systems is made by removing the three clamping foot. Allen screws (with the key for hexagon socket screws supplied), so that this feet can be turned to the particular fitting position. The milled screw must be subsequently fitted from the correct side (outer side).

This arrangement allows observation of both the longitudinal and the transverse Zeeman effect.

### 3 FUNCTIONAL AND OPERATING ELEMENTS

The Cd light source is equipped with a gas discharge lamp that is equipped with a thermionic cathode. When warmed up, a stable arc discharge burns in it. The plasma consists of an inert gas and Cd-vapour.

The lamp is held contact-safe in a housing with an integrated starter and permanently fixed connecting leads. A milled screw enables the lamp to be so mounted on the yoke of the magnet and adjusted, that the discharge tube is between the pole pieces without touching them.

The lamp can be moved sideways between the pole pieces on a guide rail so that it can be rapidly moved into and out of the magnetic field. This also allows a quick demonstration of spectral line splitting with the permanent magnet system. The stopping point of this movement can be offset by about 15 mm by unscrewing the two milled screws which hold the guide rail.

**Caution!** When using electromagnet 06480.01 and as the lamp must never be allowed to touch the pole pieces, it is essential to ensure that the pole pieces are firmly tightened so that they do not change their position under the applied electromagnet coil current. It is purposeful to check that this is so in a preliminary test without the Cd-lamp. The discharge tube is far brighter, and also far hotter, when the magnetic field is switched on. To conserve it, therefore, never keep the magnetic field switched on for longer than necessary.

Only use the lamp together with the operating unit (article no. 13662.9X\*) which is intended for this purpose.

#### 4 NOTES ON OPERATION

This high-quality instrument fulfills all of the technical requirements that are compiled in current EC guidelines. The characteristics of this product qualify it for the CE mark.

This instrument is only to be put into operation under specialist supervision in a controlled electromagnetic environment in research, educational and training facilities (schools, universities, institutes and laboratories).

A defect lamp is only to be replaced by our customer service department. Do not open the lamp housing.

#### 5 TECHNICAL SPECIFICATIONS

Operating temperature range 5...40°C

Relative humidity < 80%

|                          |                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| Supply voltage           | Only via Power supply for spectral lamps, 13662.9X*                                                       |
| Protective class         | II                                                                                                        |
| Electric power           |                                                                                                           |
| without magnetic field   | approx. 17 W                                                                                              |
| with max. magnetic field | approx. 22 W                                                                                              |
| Spectral lines used      | $\lambda = 643.8$ nm (red, normal Zeeman-Effekt)<br>$\lambda = 508.6$ nm (green, anomalous Zeeman effect) |
| Connecting lead          | $l = 110$ cm, pair of 4 mm plugs                                                                          |
| Dimensions               | (160 x 130 x 52) mm                                                                                       |
| Weight                   | approx. 0.5 kg                                                                                            |

#### 6 LITERATURE REFERENCE

|                                         |          |
|-----------------------------------------|----------|
| Handbook Laboratory Experiments Physics | 16502.32 |
| Experiment                              | P2511001 |

#### 7 LIST OF EQUIPMENT

*Main components for the Zeeman effect*

|                                 |           |
|---------------------------------|-----------|
| Fabry-Perot interferometer      | 09050.02  |
| Cadmium lamp for Zeeman effect  | 09050.20  |
| Power supply for spectral lamps | 13662.9X* |

*Magnetic systems*

|                                   |          |
|-----------------------------------|----------|
| Electromagnet without pole pieces | 06480.01 |
| Pole pieces, drilled, conical     | 06480.03 |
| Rotating table for heavy loads    | 02077.00 |

|                             |          |
|-----------------------------|----------|
| or                          |          |
| Magnetic system, adjustable | 06327.00 |

\* Voltage and frequency (see type plate) depending on local power grid.

xxxxx.91 = 115 V/60 Hz

xxxxx.92 = 115 V/50 Hz

xxxxx.94 = 230 V/60 Hz

xxxxx.97 = 230 V/50 Hz

Special voltages and fixed frequencies on request.

#### 8 NOTES ON THE GUARANTEE

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The manufacturer can only be held responsible for the function and technical safety characteristics of the instrument, when maintenance, repairs and changes to the instrument are only carried out by the manufacturer or by personnel who have been explicitly authorized by him to do so.

#### 9 WASTE DISPOSAL

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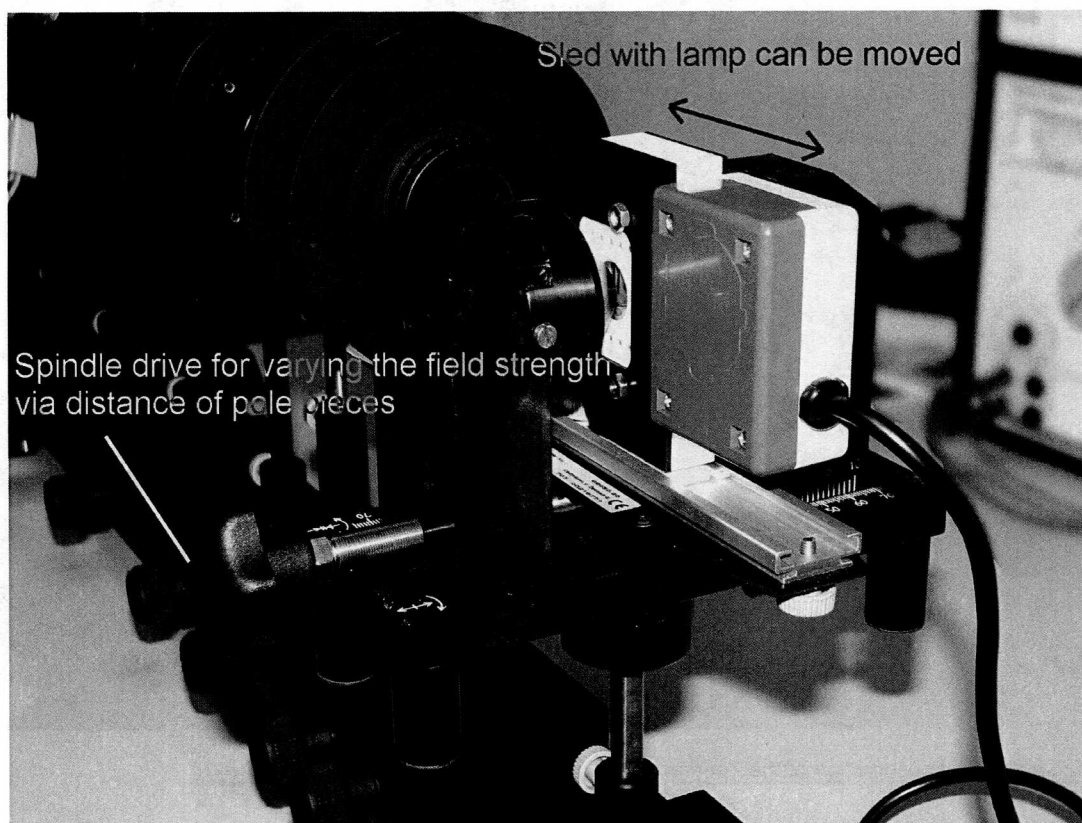
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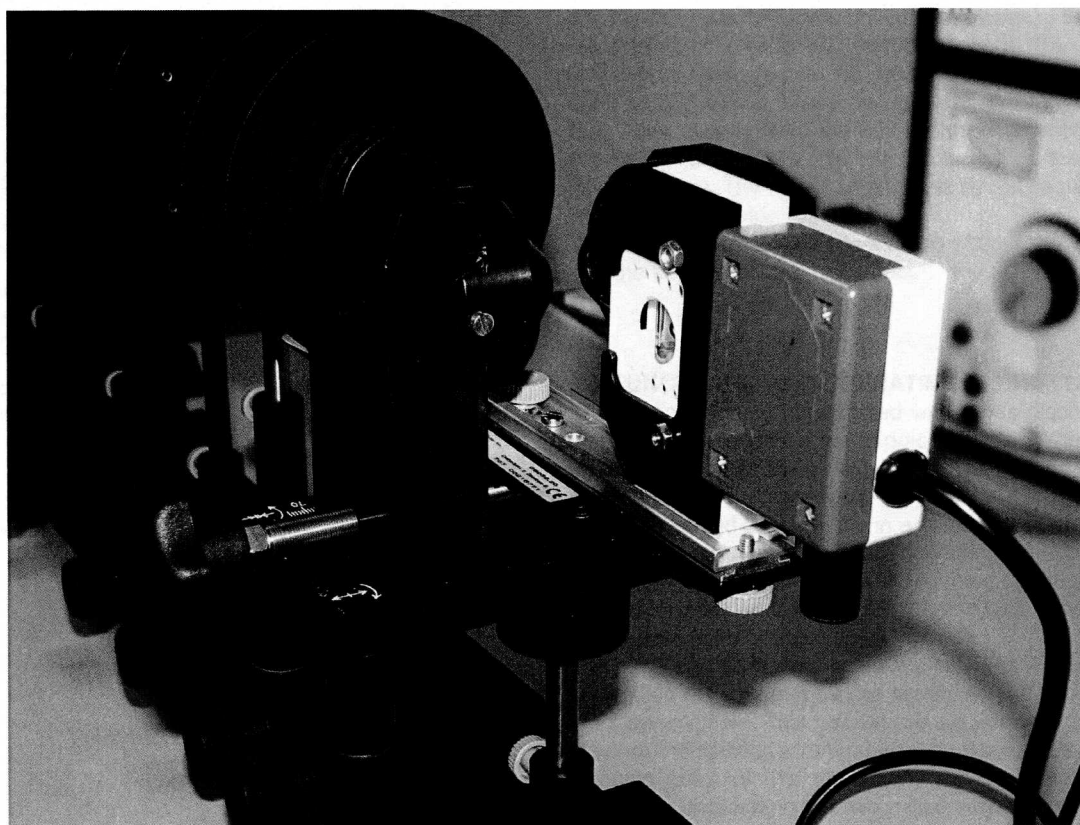
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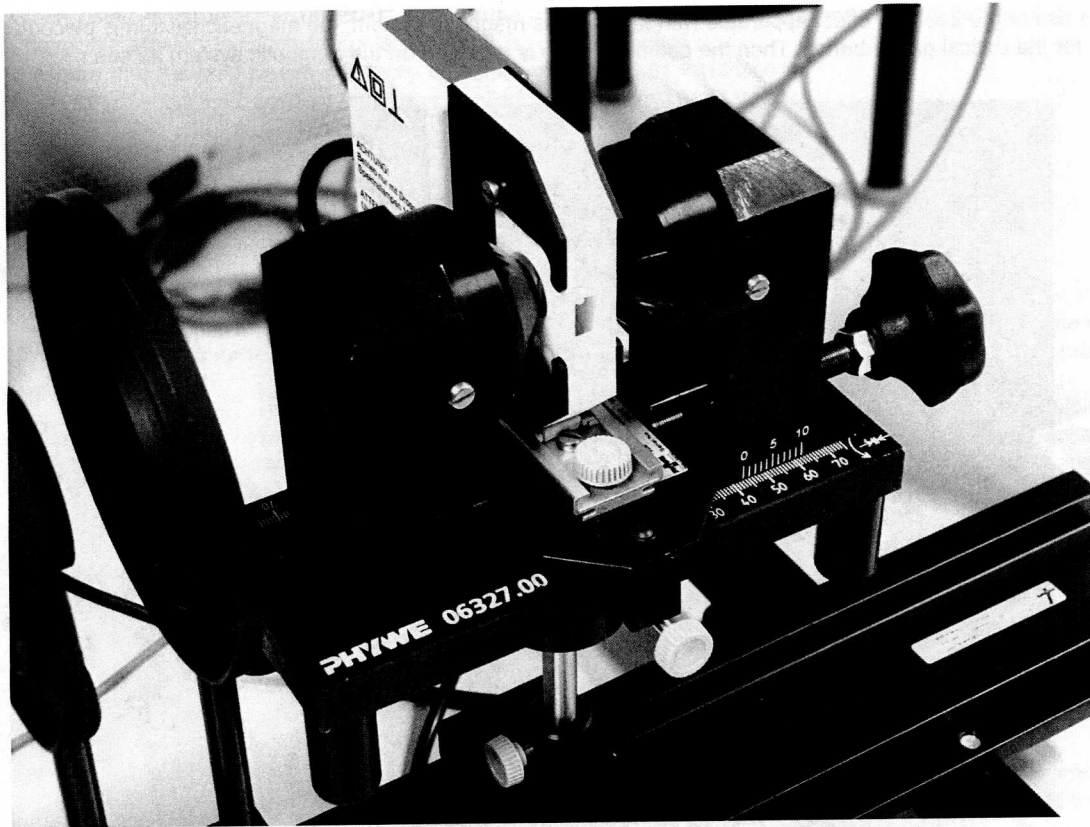


For the use of the Zeeman effect apparatus with the variable magnetic system, the magnetic system is placed on a slide mount for the optical profile bench. Then the cadmium lamp is mounted on the magnetic system as seen.



In transversal observation of the discharge the magnetic field strength can be varied by either moving the lamp in and out of the field or by varying the spacing between the magnets.





Here the longitudinal observation of the Zeeman effect can be seen.



Var. Transformer, 25 VAC/20 VDC, 12 A  
Var. Transformer, 25 VAC/20 VDC, 12 A,  
with analogue displays

13531.90... 99  
13532.90... 99

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## Operating instructions



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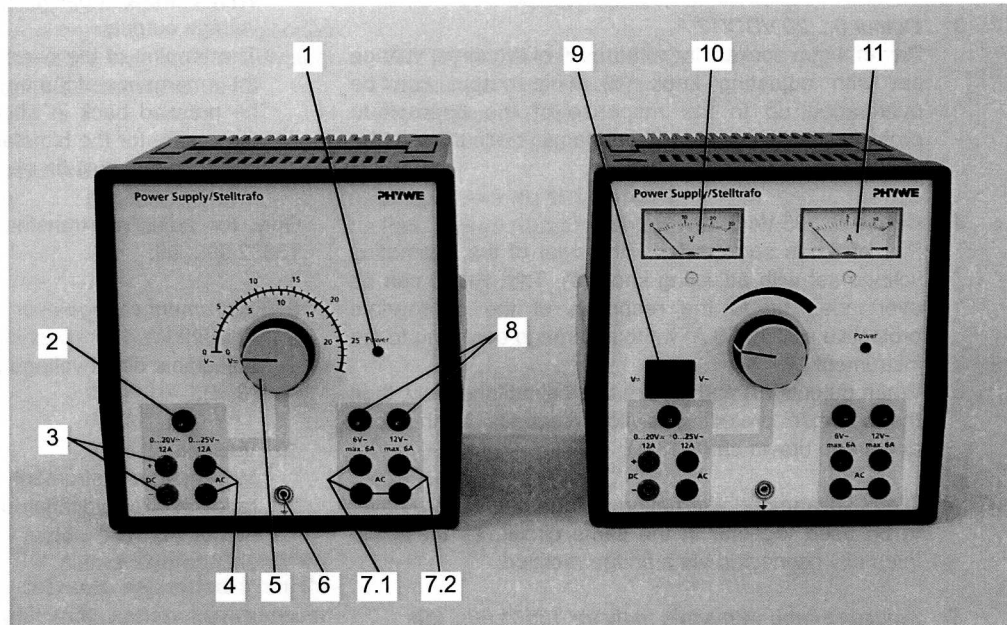


Fig. 1: Front view of the Var. Transformer 25 VAC/20 VDC, 12 A 13531.93 and 13532.93.

## 1 SAFETY PRECAUTIONS



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- Check that your mains supply voltage corresponds to that given on the type plate fixed to the instrument.
- Install the instrument so that the on/off switch and the mains connecting plug are easily accessible.
- Do not cover the ventilation slots.
- Take care that no liquids or objects enter in through the ventilation slots.
- Only use the instrument in dry rooms in which there is no risk of explosion.
- Do not start up this instrument in case of visible signs of damage to it or to the line cord.
- Only use the instrument for the purpose for which it was designed.

## 2 PURPOSE AND DESCRIPTION

The practical power supply unit provides two almost continuously adjustable low voltages, one with a 100 Hz pulsating direct voltage 0... 20 VDC/12 A and the other with an alternating voltage of 0... 25 VAC/12 A. In addition to these, two fixed voltages of 6 VAC/max. 6 A and 12 VAC/max. 6 A are also available.

All outputs are galvanic separated from the mains, not earthed and protected by an overcurrent circuit breaker.

The 1.5 m long connecting cord that is supplied serves to connect the adjustable transformer to the alternating current mains. For this purpose it is fitted onto the connecting plug at the back of the instrument. The square fuse holder built-in below the instrument connecting plug can only be opened with a screw driver or similar when the connecting cord has been disconnected from the transformer.

Both conventional 4 mm plugs and also safety connecting cords (e.g. 07337.01) fit into the safety output sockets.

## 3 FUNCTIONAL AND OPERATING ELEMENTS

The adjustable transformer is accommodated in a shock-resistant plastic housing. A carrying handle that can be turned upwards is recessed in the cover plate of the instrument housing. A similar handle is recessed in the bottom plate, but this can be turned to the back to enable the instrument to be set up in an inclined position. Four rubber feet guard against slipping and ensure stability. The adjustable transformer can be stacked on other instruments of the same design in the inclined position, whereby the rubber feet fit in pan-shaped recesses in the top cover of the instrument below it for further resistance to slipping. In stacks of such instruments, it is only permissible to have the one at the top in the inclined position.

The following functional and operating elements (see Fig. 1) are to be found on the front plate of the instrument:

1 *On/off indicating lamp*

2 *Overcurrent protective switch*

with thermal tripping for fuse protection of outputs (3) and (4).

3 *Output 0... 20 VDC/12 A*

Pair of 4 mm sockets for withdrawal of the direct voltage set with adjusting knob (5). This output can be overloaded up to the response of the appropriate protective switch (13 A) without damage occurring to the instrument.

4 *Output 0... 25 VAC/12 A*

Pair of 4 mm sockets for withdrawal of the alternating voltage set with adjusting knob (5). This output can be overloaded up to the response of the appropriate protective switch (13 A) without damage occurring to the instrument.

When outputs (3) and (4) are used simultaneously, then the common protective switch responds to the total amperage drawn off (> 13 A).

Direct voltage and alternating voltage outputs are never to be used together in the same circuit, as these are internally connected via a bridge rectifier!

5 *Adjusting knob with scale (only for 13531.90... 99)*

This allows the required voltages to be set at outputs (3) and (4), whereby the two scales allow rapid orientation. At a given setting, the height of the voltage is dependent to a certain extent on the load; the scale values are valid at approximately half the nominal current (6 A) at a nominal value of the mains voltage of 230 VAC. With lower loads, the output voltage is less. A voltmeter must be used for an exact determination of the set voltages.

6 *Earth socket*

On connection to the mains, this is connected to the protective earthing conductor.

7 *Outputs 6 VAC and 12 VAC*

Two 4 mm pairs of sockets for withdrawal of the fixed alternating voltages, i.e. a voltage of 6 VAC/max. 6 A at (7.1) and 12 VAC/max. total current 6 A at (7.2).

These two output voltages can so not be switched in series by connecting the upper socket of the one output with the lower socket of the other output. This would rather result in a short circuit, against which, however, the instrument is fuse protected.

The height of the voltage at outputs (7.1) and (7.2) depends to a certain extent on the load. The 12 V nominal voltage is given at the mains voltage nominal value of 230 VAC and at half the nominal current (3 A). At the 6 VAC output, the nominal value is already given at a current of approx. 1.7 A, so that the frequently used 6 V/10 W halogen incandescent lamps, that are sensitive to overvoltage, are not endangered.

Outputs (7.1) and (7.2) are each protected by a 6 A overcurrent protective switch. When only one output is used, then this can be - independently of a possible loading of outputs (3) and (4) - overloaded up to the response of this protective switch without damage occurring to the instrument. Only the output voltage drops correspondingly with the higher load. Should

outputs (7.1) and (7.2) be simultaneously used, however, then care must be taken that the total current intensity withdrawn is limited to a maximum of 6 A.

The fixed voltage outputs are suitable, for example, as voltage suppliers to lamps with 6 V or 12 V nominal voltage and a nominal current intensity of up to 6 A.

8 *Overcurrent protective switch*

With thermal tripping, for fuse protection of the fixed voltage outputs.

The tripping of the overcurrent protective switch effects the emergence of the appropriate safety button; this can be pressed back in after a some seconds, which are necessary for the bimetal to be able to cool. The cause of the fault should be eliminated in the meantime.

Only for adjusting transformers with analogue displays, 13532.90... 99:

9 *Instrument change-over switch*

This allows the switching over of the voltmeter to the adjustable direct voltage or alternating voltage output (3, 4).

10 *Voltmeter*

Moving iron instrument for the display of the voltage ready at direct voltage output (3) or at alternating voltage output (4): The output is selected with the instrument change-over switch.

11 *Ammeter*

Moving iron instrument for the display of the current withdrawn at direct voltage output (3) or at alternating voltage output (4). (The ammeter shows the total current of the two outputs).

The display instruments are moving iron instruments. These instruments show the correct effective value, not only in alternating voltage measurements but also in measurements of pulsating direct voltage and direct current at output (1). In contrast, customary moving-coil instruments with rectifier measure the mean value, which is about 11% lower than the effective value at output (1).



**Caution!**

The power supply is to be exclusively used for the supply of appropriate electrical experimental set-ups and instruments. The user carries the responsibility for the operational safety of set-ups that he or she connects to it. Should incorrect circuits be connected, then even the relatively low performance provided by the power supply can cause considerable damage (fire hazard!). To avoid unnecessary risks, it is therefore recommended that you carefully check the set-up that is to be supplied to prior to switching the power supply on.



#### 4 NOTES ON OPERATION

This high-quality instrument fulfills all of the technical requirements that are compiled in current EC guidelines. The characteristics of this product qualify it for the CE mark.

This instrument is only to be put into operation under specialist supervision in a controlled electromagnetic environment in research, educational and training facilities (schools, universities, institutes and laboratories).

This means that in such an environment, no mobile phones etc. are to be used in the immediate vicinity. The individual connecting leads are each not to be longer than 2 m.

#### 5 TECHNICAL SPECIFICATIONS (typical for 25°C)

Operating temperature range 5... 40°C  
Relative humidity < 80%

##### Mains supply

Protection class I  
Connecting voltage see type plate  
(+6%/-10%)  
Mains frequency 50/60 Hz  
Power consumption approx. 375 VA  
Mains fuse see type plate  
(5 mm x 20 mm)

##### Outputs

**Galvanic separated from the mains and not earthed, fixed voltage outputs also galvanic separated from the adjustable outputs. No galvanic separation between adjustable direct voltage and alternating voltage outputs!**

Fixed voltage 6/12 V~  
Nominal current max. 6 A  
(total current for both outputs)  
Overload protection Overcurrent circuit breaker

Adjustable voltages  
Alternating voltage 0... 25 V~  
Direct voltage 0... 20 V-  
Ripple 48%  
Nominal current max. 13 A  
(total current for both outputs)  
Overload protection Overcurrent circuit breaker

Housing dimensions (mm) 230 x 236 x 250 (W, D, H)  
Weight approx. 10 kg

#### 6 NOTES ON THE GUARANTEE

We guarantee the instrument supplied by us for a period of 24 months within the EU, or for 12 months outside of the EU. This guarantee does not cover natural wear nor damage resulting from improper handling.

The manufacturer can only be held responsible for the function and technical safety characteristics of the instrument, when maintenance, repairs and changes to the instrument are only carried out by the manufacturer or by personnel who have been explicitly authorized by him to do so.

#### 7 WASTE DISPOSAL

The packaging consists predominately of environmentally compatible materials that can be passed on for disposal by the local recycling service.



Should you no longer require this product, do not dispose of it with the household refuse. Please return it to the address below for proper waste disposal.

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D-37079 Göttingen

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